SIMATS ENGINEERING

Assessment Tool 3 – Coding Challenge

Course: Embedded Systems (ECA1406) | CO2 – BL5

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Question No: 28

Title: Design of an Embedded C Firmware for Fault Detection Using Threshold Logic in Industrial Systems

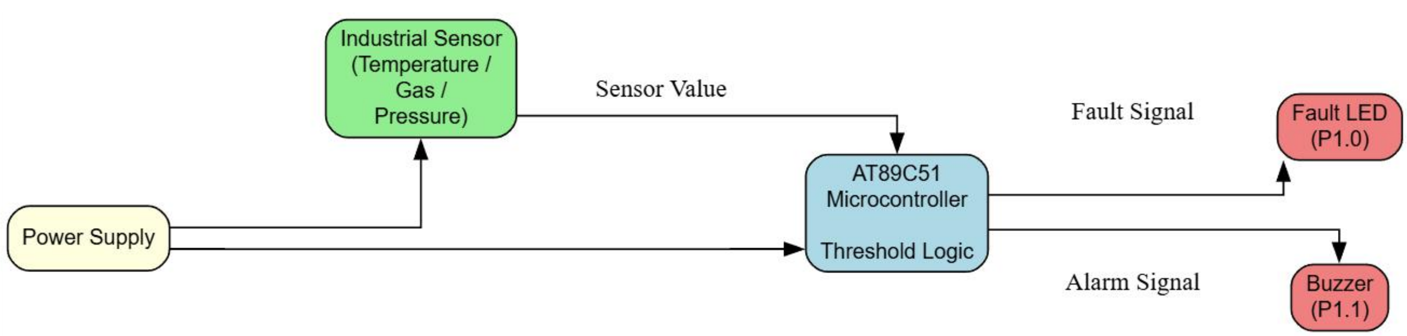
1. Objective

To design and implement an Embedded C firmware for the AT89C51 microcontroller that continuously monitors a simulated industrial sensor value, compares it with a predefined threshold, and activates a fault LED and buzzer whenever the threshold is exceeded. The program demonstrates threshold-based fault detection commonly used in industrial automation and safety systems.

2. Problem Statement

Design an Embedded C firmware for fault detection using threshold logic in industrial systems. The program should continuously monitor the sensor value, compare it with a predefined threshold, and generate an alarm whenever the measured value exceeds the safe operating limit. This ensures timely fault detection and improves system reliability in industrial environments.

3. Block Diagram



4. **Algorithm**

- Start the program.
- Configure P1.0 as the Fault LED output.
- Configure P1.1 as the Buzzer output.
- Initialize the threshold value (threshold = 50).
- Read the sensor value using ReadSensor().
- Compare the sensor value with the threshold in CheckFault().
- If the sensor value is greater than the threshold: turn ON the Fault LED and Buzzer.
- Otherwise: turn OFF the Fault LED and Buzzer.
- Repeat the process continuously using an infinite while(1) loop.

5. **Coding Approach**

The firmware is developed using Embedded C for the AT89C51 microcontroller with the reg51.h header file. A simulated sensor value is used to represent an industrial parameter such as temperature, pressure, or gas concentration. The firmware continuously compares the sensor value with a predefined threshold. When the measured value exceeds the threshold, the microcontroller activates a Fault LED connected to Port 1.0 and a buzzer connected to Port 1.1. If the sensor value remains within the safe limit, both outputs remain OFF. This threshold-based monitoring technique is widely used in industrial automation to provide early warning of abnormal operating conditions. The logic is split into two modular functions — ReadSensor() and CheckFault() — called repeatedly from main(), keeping the design simple, portable, and easy to extend to real ADC-based sensor input.

6. Source Code

```
#include <reg51.h>
sbit FAULT_LED = P1^0;
sbit BUZZER    = P1^1;
unsigned char sensorValue = 0;
unsigned char threshold = 50;
void ReadSensor(void)
```

```
{
    sensorValue = 60;
}
void CheckFault(void)
{
    if(sensorValue > threshold)
    {
        FAULT_LED = 1;
        BUZZER = 1;
    }
    else
    {
        FAULT_LED = 0;
        BUZZER = 0;
    }
}
void main(void)
{
    while(1)
    {
        ReadSensor();
        CheckFault();
    }
}
```

7. Output / Screenshots

7.1 Source Code and Successful Build in Keil µVision

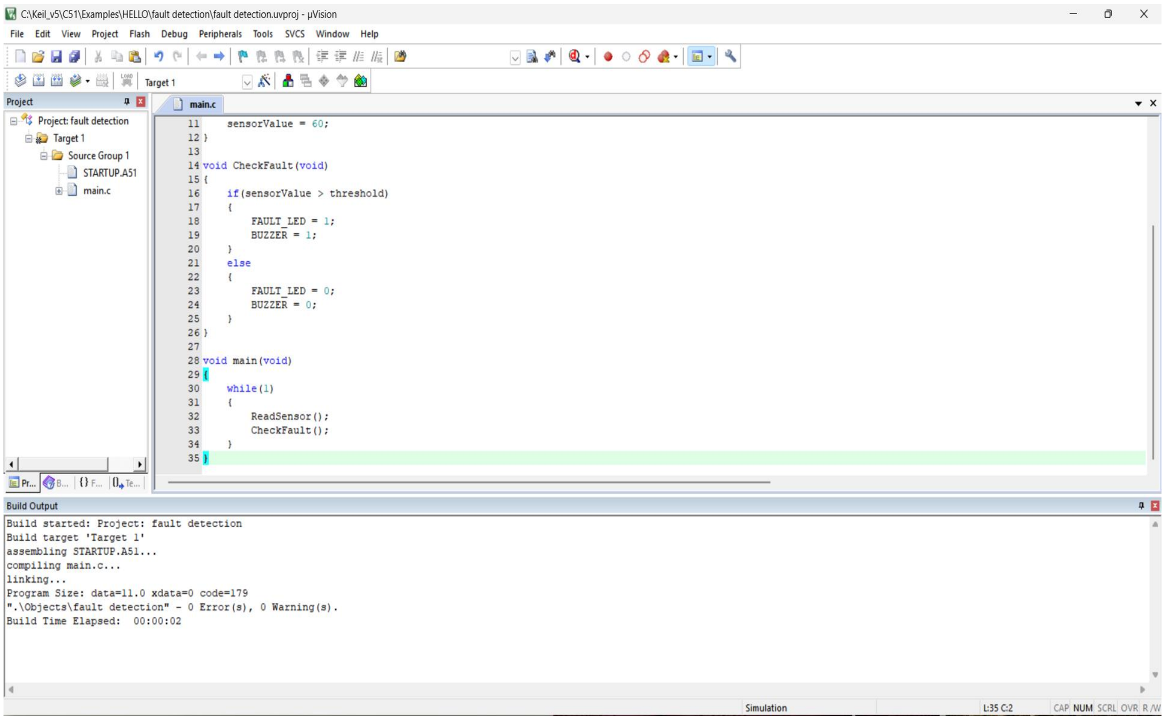


Fig 1: CheckFault() source code with successful build output — 0 Errors, 0 Warnings

7.2 Debug Execution

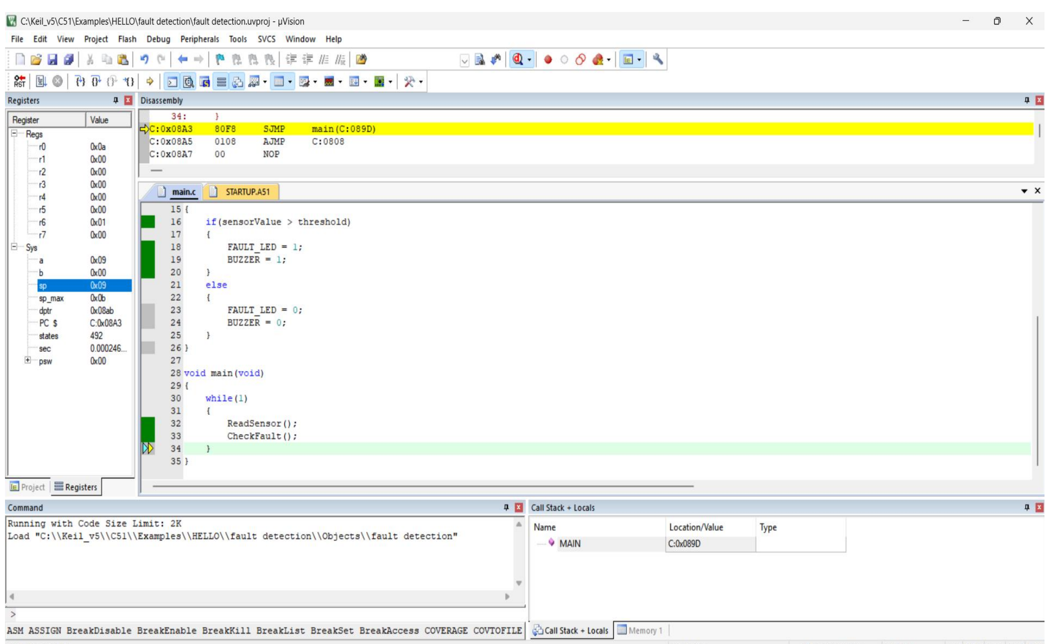


Fig 2: Debug session showing execution inside CheckFault(), FAULT_LED and BUZZER set to 1 (R6 = 0x01 in Registers)

7.3 Output Validation

- Sensor Value = 60
- Threshold = 50
- Result: Fault Detected
- Fault LED = ON
- Buzzer = ON

Since the sensor value (60) is greater than the threshold (50), the firmware successfully detects the fault and activates both the LED and buzzer, as confirmed in the debug session above.

8. Competitive Performance Analysis

Parameter	Observation
Sensor Monitoring	Continuous monitoring using Embedded C
Threshold Comparison	Compares sensor value with predefined limit
Fault Detection	Detects abnormal conditions immediately
Fault Indication	Activates LED and buzzer during fault
CPU Utilization	Simple continuous polling approach
Industrial Suitability	Suitable for basic industrial safety monitoring applications

9. Conclusion

An Embedded C firmware for industrial fault detection using threshold logic was successfully designed and implemented on the AT89C51 microcontroller. The program continuously monitors a simulated sensor value, compares it with a predefined threshold, and generates an alarm whenever a fault condition is detected. The firmware correctly activates the Fault LED and buzzer when the sensor value exceeds the safe operating limit. This approach demonstrates a simple and effective fault detection mechanism suitable for industrial monitoring and safety applications.